

Can $(1 - \cos)^3$ Replace tEDS as a DM-Triggered Early Dark Energy Potential?

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In the Early Dark Sector framework, a coupling between the early dark energy (EDE) scalar field and interacting dark matter (IDM) can trigger the EDE release near matter-radiation equality, dynamically resolving the coincidence problem. This mechanism has been demonstrated for the tEDS plateau potential, which features a broad flat region that suppresses the bare slope and allows the coupling to dominate the onset of field motion. We ask whether the standard axion-like $(1 - \cos \theta)^3$ hilltop potential can achieve the same coupling-triggered release. We show numerically that for the quadratic IDM mass coupling $m_{\text{IDM}}(\phi) = m_0(1 + g\phi^2)$ with $g \sim 0.5$ – 1 , the coupling force completely dominates the onset of field motion for *all* initial conditions tested, including θ_i far from the hilltop. The practical difference between the two potentials is not a fine-tuning requirement but a correlation: in $(1 - \cos)^3$, the initial field offset $\delta \equiv \pi - \theta_i$ simultaneously controls the amplitude (f_{EDE}) and the release timing (z_{peak}) because larger δ implies earlier release into a more radiation-dominated background. For $g \simeq 0.68$ and $\delta \in [0.1, 1]$, we find $f_{\text{EDE}} \in [0.06, 0.11]$ with z_{peak} spanning the equality scale, without requiring θ_i to be tuned to within a percent of π . The main limitation of $(1 - \cos)^3$ relative to tEDS is the amplitude–timing degeneracy: tEDS has an independent normalization V_0 that decouples the EDE height from the trigger redshift, while in $(1 - \cos)^3$ these are tied through δ and m .

I. INTRODUCTION

Early dark energy (EDE) reduces the Hubble tension by injecting extra energy density around matter-radiation equality, shortening the pre-recombination sound horizon while preserving the observed CMB angular scale [1, 2]. A canonical EDE realization uses a scalar field initially held near a hilltop by Hubble friction, with the release triggered when the field starts to roll. A well-studied potential is the axion-like

$$V(\phi) = m^2 f^2 \left[1 - \cos\left(\frac{\phi}{f}\right) \right]^3, \quad (1)$$

which, near the hilltop $\theta_i = \phi_i/f \approx \pi$, holds the field in a high-energy state until the Hubble friction decreases sufficiently to allow rolling.

In this uncoupled picture, the release timing depends on the bare competition between m , f , and H , requiring m to be tuned to the equality scale—a coincidence problem. The Early Dark Sector (EDS) framework of Ref. [3] resolves this by introducing a coupling between the EDE scalar field and a species of interacting dark matter (IDM) whose mass depends on the field,

$$m_{\text{IDM}}(\phi) = m_0(1 + g\phi^2), \quad (2)$$

producing an extra force in the Klein-Gordon equation,

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} + 3\rho_{\text{IDM}}\partial_{\phi}\ln m_{\text{IDM}} = 0. \quad (3)$$

Since $\rho_{\text{IDM}} \propto a^{-3}$ scales like matter, while $H^2 \propto \rho_{\text{tot}}$ drops faster in the radiation era, the trigger condition $|3\rho_{\text{IDM}}\partial_{\phi}\ln m_{\text{IDM}}| \sim H^2$ is satisfied near matter-radiation equality independently of the potential shape. This is the mechanism studied in Ref. [4] using the tEDS plateau potential

$$V(\phi) = V_0 \frac{\theta^6}{(1 + \theta^{6p})^{1/p}}, \quad \theta \equiv \phi/f_{\text{tEDS}}, \quad (4)$$

which has a broad plateau at large θ where $V_{,\phi}$ is parametrically suppressed.

The key question we address here is: *does the $(1 - \cos)^3$ potential work equally well as a DM-triggered EDE model, and if so what is the physical difference from tEDS?* We implement both potentials in AxiCLASS with the type-1 coupling Eq. (2) and perform a systematic background-level comparison.

II. TRIGGER MECHANISM AND DIAGNOSTICS

A. Overdamped branch

At early times the field follows the overdamped terminal branch (set $\ddot{\phi} \approx 0$):

$$\dot{\phi}_{\text{term}} \simeq -\frac{V_{,\phi} + 3\rho_{\text{IDM}} \partial_{\phi} \ln m_{\text{IDM}}}{3H}. \quad (5)$$

Release departure from slow roll occurs when Eq. (5) breaks down, i.e., when the effective force $\mathcal{F} \equiv V_{,\phi} + 3\rho_{\text{IDM}} \partial_{\phi} \ln m_{\text{IDM}}$ becomes comparable to $3H\dot{\phi}$.

B. Diagnostics

We define two diagnostics. The *coupling fraction*

$$f_{\text{coup}}(z) \equiv \frac{|3\rho_{\text{IDM}} \partial_{\phi} \ln m_{\text{IDM}}|}{|V_{,\phi}| + |3\rho_{\text{IDM}} \partial_{\phi} \ln m_{\text{IDM}}|} \quad (6)$$

measures the instantaneous share of the total force carried by the coupling, ranging from 0 (bare-slope dominated) to 1 (coupling dominated). We evaluate it at two moments:

- *at the trigger*: the first redshift z_{trig} at which $|\phi - \phi_i|$ exceeds 5% of $|\phi_i|$;
- *at the peak*: the redshift z_{peak} at which $f_{\text{EDE}}(z)$ is maximum.

The trigger value $f_{\text{coup}}(z_{\text{trig}})$ measures whether the coupling initiates the rolling. The peak value $f_{\text{coup}}(z_{\text{peak}})$ measures whether the coupling continues to dominate throughout the release. These two quantities carry qualitatively different information for the two potentials, as we now show.

III. RESULTS

A. Both potentials are coupling-triggered at onset

Figure 8 shows $f_{\text{coup}}(z)$ as a function of redshift for tEDS (left) and $(1 - \cos)^3$ (right), with the trigger redshift marked by a vertical dotted line for each run. The key result is:

$$f_{\text{coup}}(z_{\text{trig}}) = 1.00 \quad \text{for all cases in both potentials.} \quad (7)$$

The coupling completely dominates the onset of field motion in both models. This confirms that both potentials implement the same coupling-triggered release mechanism when $g \gtrsim 0.5$.

B. Different post-trigger behavior

Despite the identical trigger coupling fractions, the two potentials differ sharply in the post-trigger evolution:

a. tEDS. After the field is pushed off the plateau ($z_{\text{trig}} \sim 10^5$), it rolls rapidly off the steep wall of the tEDS potential toward $\phi \rightarrow 0$. At the EDE peak, $\phi \approx 0$, so $\partial_{\phi} \ln m_{\text{IDM}} = 2g\phi/(1 + g\phi^2) \rightarrow 0$ and the coupling force vanishes. By the time f_{EDE} reaches its maximum, $f_{\text{coup}}(z_{\text{peak}}) \simeq 0$ and the release is driven by the bare potential slope.

b. $(1 - \cos)^3$. The field starts near $\theta_i \approx \pi$, corresponding to $\phi_i \approx \pi f$ (large for $f \sim 1 M_{\text{Pl}}$). The coupling force $\partial_{\phi} \ln m_{\text{IDM}} = 2g\phi/(1 + g\phi^2)$ remains of order unity at $\phi \sim \pi f$. Because the bare slope $V_{,\phi} \sim 12m^2 f(\pi - \theta)$ is small near the hilltop, the coupling retains $f_{\text{coup}} \approx 1$ throughout the release and even at the peak. The coupling is not merely a trigger—it continuously drives the terminal velocity during the entire EDE episode.

The force budget is illustrated in Fig. 9: for $g = 0.68$ and any $\delta \equiv \pi - \theta_i$ in the range $[0.01, 2]$, the coupling term exceeds the bare slope by several orders of magnitude at the trigger redshift, and the hierarchy is maintained until well after the peak.

C. Release redshift and amplitude: the amplitude–timing correlation

Figure 6 shows the release redshift z_{peak} and peak amplitude $f_{\text{EDE}}^{\text{peak}}$ as a function of δ for each value of g . Two features are evident:

1. *Later release for smaller δ .* The bare slope $V_{,\phi} \approx 12m^2f\delta$ is smaller for smaller δ , so the coupling must work longer (the IDM density must dilute further) before the effective force is large enough to initiate rolling. Smaller δ therefore shifts z_{peak} to lower redshift (later release).
2. *Larger amplitude for smaller δ .* The peak EDE fraction is $f_{\text{EDE}}^{\text{peak}} \simeq \rho_{\phi}(z_{\text{peak}})/\rho_{\text{tot}}(z_{\text{peak}})$. Since $\rho_{\phi} \approx V(\theta_i) \approx 8m^2f^2$ is nearly independent of δ for $\delta \ll 1$ (all initial conditions near the hilltop have the same potential energy), while ρ_{tot} is *smaller* at lower z (less radiation), a later release produces a larger $f_{\text{EDE}}^{\text{peak}}$.

These two effects conspire: δ controls both the timing and the amplitude, and they move in opposite directions. Increasing δ shifts the release earlier and reduces $f_{\text{EDE}}^{\text{peak}}$; decreasing δ shifts the release later and increases $f_{\text{EDE}}^{\text{peak}}$.

For $g = 0.68$, the window $\delta \in [0.1, 0.5]$ gives $f_{\text{EDE}}^{\text{peak}} \in [0.09, 0.11]$ and $z_{\text{peak}} \in [2300, 2900]$, which brackets the standard EDE target of $f_{\text{EDE}} \sim 0.1$ at $z \sim z_{\text{eq}}$. No extreme fine-tuning of θ_i toward π is required. For $g = 1.0$, the window shifts to $\delta \in [0.5, 1]$ giving $f_{\text{EDE}} \in [0.08, 0.11]$ and $z_{\text{peak}} \in [3600, 5200]$, somewhat closer to or above the equality scale.

D. Comparison of histories

Figure 7 shows $f_{\text{EDE}}(z)$ and $\theta(z)$ for tEDS (left) and $(1 - \cos)^3$ (right). The tEDS histories show a sharp EDE bump followed by rapid decay: the field rolls off the plateau quickly, overshoots to small ϕ , and decays. The $(1 - \cos)^3$ histories show a broader, lower bump: the field drifts away from the hilltop more gradually, and the bump is wider in $\log z$.

The width of the EDE bump has direct observational consequences. A broader bump integrates more energy and requires a larger effective f_{EDE} at equality to shift H_0 by the same amount as a sharp bump. This may require somewhat larger coupling in the $(1 - \cos)^3$ case to reach the same observational signal as tEDS.

IV. KEY DIFFERENCE: AMPLITUDE–TIMING DEGENERACY

The comparison above identifies one fundamental structural difference between the two potentials:

- **tEDS:** the normalization V_0 controls the EDE amplitude independently of the trigger timing, which is set by θ_i and g . One can in principle independently dial the height of the f_{EDE} bump and the redshift at which it peaks.
- $(1 - \cos)^3$: the amplitude and timing are both controlled by $\delta \equiv \pi - \theta_i$ (and jointly by m, f). Increasing the amplitude (smaller δ) delays the release, and vice versa. The product $m \cdot f$ sets an overall energy scale, while δ and g together determine the timing. This leaves less independent freedom to optimize both the height and the position of the EDE bump.

This degeneracy is the main practical limitation of $(1 - \cos)^3$ compared to tEDS for the Hubble tension. When fitting CMB data, the optimal $f_{\text{EDE}}^{\text{peak}}$ and z_{peak} are set by the CMB likelihood, and a model with two coupled parameters instead of two independent parameters has a smaller effective parameter space.

V. SUMMARY

We have shown that the $(1 - \cos)^3$ hilltop potential *can* operate as a DM-triggered EDE model: with $g \gtrsim 0.5$, the coupling completely dominates the onset of field motion ($f_{\text{coup}}(z_{\text{trig}}) = 1$) and sets the release redshift near equality for $\delta \in [0.1, 1]$, without requiring extreme initial-condition tuning.

The physical difference from tEDS is not about fine-tuning but about the structure of the parameter space:

1. In tEDS, the post-trigger dynamics are dominated by the bare slope (the field overshoots to $\phi \rightarrow 0$), while in $(1 - \cos)^3$ the coupling remains active throughout ($f_{\text{coup}} \approx 1$ at the peak).
2. In tEDS, V_0 and the trigger parameters (g, θ_i) are largely independent. In $(1 - \cos)^3$, the amplitude and timing are coupled through δ , reducing the effective parameter freedom.

These properties make tEDS a more flexible construction for the Hubble tension phenomenology, but they do not prevent $(1 - \cos)^3$ from achieving coupling-triggered release. A full CMB-level fit would be needed to determine whether the amplitude–timing coupling is a significant disadvantage compared to the prior freedom in δ .

Appendix A: Reproduction of Figure 2 of Ref. [4]

Figure 2 of Ref. [4] is the central numerical demonstration of the tEDS trigger mechanism. It shows $\theta(z)$, the fractional DM mass shift $\Delta m/m = g\phi^2$, and $f_{\text{EDE}}(z)$ for three initial conditions $\theta_i \in \{4, 6, 8\}$ on the tEDS plateau ($p = 8$, $f_{\text{tEDS}} = 0.05 M_{\text{Pl}}$), each with a slightly different coupling $g \in \{0.56, 0.68, 0.75\}$ tuned to achieve comparable f_{EDE} peaks. The key feature is that all three cases release at the *same* redshift despite spanning a factor of two in θ_i , demonstrating that the trigger timing is set by the IDM dilution rate and not by the initial field position.

Our AxiCLASS reproduction is shown in Fig. 1. The three panels reproduce the same qualitative structure as the original figure.

a. Agreement. With the corrected IDM normalization (CLASS internal shooting enforcing $\omega_{\text{IDM},0} = 0.1280 h^2$, see Appendix C), we find $\log_{10} z_{\text{peak}} = 3.53$ ($z_{\text{peak}} \approx 3380$) for all three cases, compared to the paper’s $z_c \approx 3162$ ($\log_{10} \approx 3.50$). The agreement is excellent: the $\sim 7\%$ offset in z_{peak} is consistent with our higher $H_0 = 72.81 \text{ km/s/Mpc}$ relative to the paper’s best-fit $H_0 \approx 70.7 \text{ km/s/Mpc}$. The most important feature – degenerate peak redshift across all θ_i – is confirmed. The DM mass shifts $g\phi_i^2 = 0.12, 0.061, 0.022$ for $\theta_i = 8, 6, 4$ match the ordering in the original figure.

b. Residual discrepancy. The small remaining offset $\Delta \log_{10} z_{\text{peak}} \approx 0.03$ (ours above the paper) is most likely an H_0 effect: a lower H_0 shifts matter–radiation equality to slightly lower z , pulling the trigger down. The f_{EDE} amplitudes (≈ 0.113 for all three) are slightly above the paper’s $\approx 10\%$; the residual difference is discussed in the discrepancy paragraph. The key qualitative result – coupling-dominated, θ_i -independent release – is correctly reproduced.

Appendix B: Why f_{coup} stays large in $(1 - \cos)^3$ but collapses in tEDS

The coupling fraction $f_{\text{coup}}(z)$ remains near unity throughout the $(1 - \cos)^3$ release but collapses to ≈ 0 at the tEDS peak. Both behaviors follow from the same polynomial coupling $\partial_\phi \ln m_{\text{IDM}} = 2g\phi/(1 + g\phi^2)$ combined with the very different field ranges involved.

For tEDS ($f_{\text{tEDS}} = 0.05 M_{\text{Pl}}$), the field starts at $\phi_i = \theta_i f_{\text{tEDS}} = 0.3 M_{\text{Pl}}$ for $\theta_i = 6$. After the plateau the potential has a steep wall, and the field rolls rapidly to $\phi \approx 0$ (confirmed numerically: $\phi_{\text{peak}} = 0.041 M_{\text{Pl}}$, $\phi_{\text{final}} \approx 0$). At ϕ_{peak} :

$$\partial_\phi \ln m_{\text{IDM}}|_{\text{tEDS, peak}} = \frac{2(0.68)(0.041)}{1 + (0.68)(0.041)^2} \approx 0.056. \quad (\text{B1})$$

The coupling force $3\rho_{\text{IDM}}\partial_\phi \ln m_{\text{IDM}}$ is negligible compared to $V_{,\phi}$ at the steep wall, so $f_{\text{coup}} \rightarrow 0$.

For $(1 - \cos)^3$ ($f = 1 M_{\text{Pl}}$), the field starts at $\phi_i = \theta_i f \approx \pi M_{\text{Pl}} \approx 3.04 M_{\text{Pl}}$. The potential near the hilltop is flat (saddle point with $V'' < 0$), and the field rolls slowly. Numerically, $\phi_{\text{peak}} \approx 1.0 M_{\text{Pl}}$ for $\delta = 0.1$ – the field has moved from $\theta \approx \pi$ to $\theta \approx 1$ by peak time, but ϕ is still at the Planck scale. At ϕ_{peak} :

$$\partial_\phi \ln m_{\text{IDM}}|_{\text{cos}^3, \text{peak}} = \frac{2(0.68)(1.0)}{1 + (0.68)(1.0)^2} \approx 0.81. \quad (\text{B2})$$

The coupling force is large because ϕ is still large. Combined with $V_{,\phi} \propto (\pi - \theta)$ being modest near the hilltop, f_{coup} remains close to unity.

The root cause is the ratio of field scales: $f/f_{\text{tEDS}} = 20$. The polynomial coupling $\partial_\phi \ln m_{\text{IDM}} \propto \phi$ is naturally small when $\phi \ll M_{\text{Pl}}$ (as in tEDS where $\phi \leq 0.4 M_{\text{Pl}}$) and naturally large when $\phi \sim M_{\text{Pl}}$ (as in the $(1 - \cos)^3$ case).

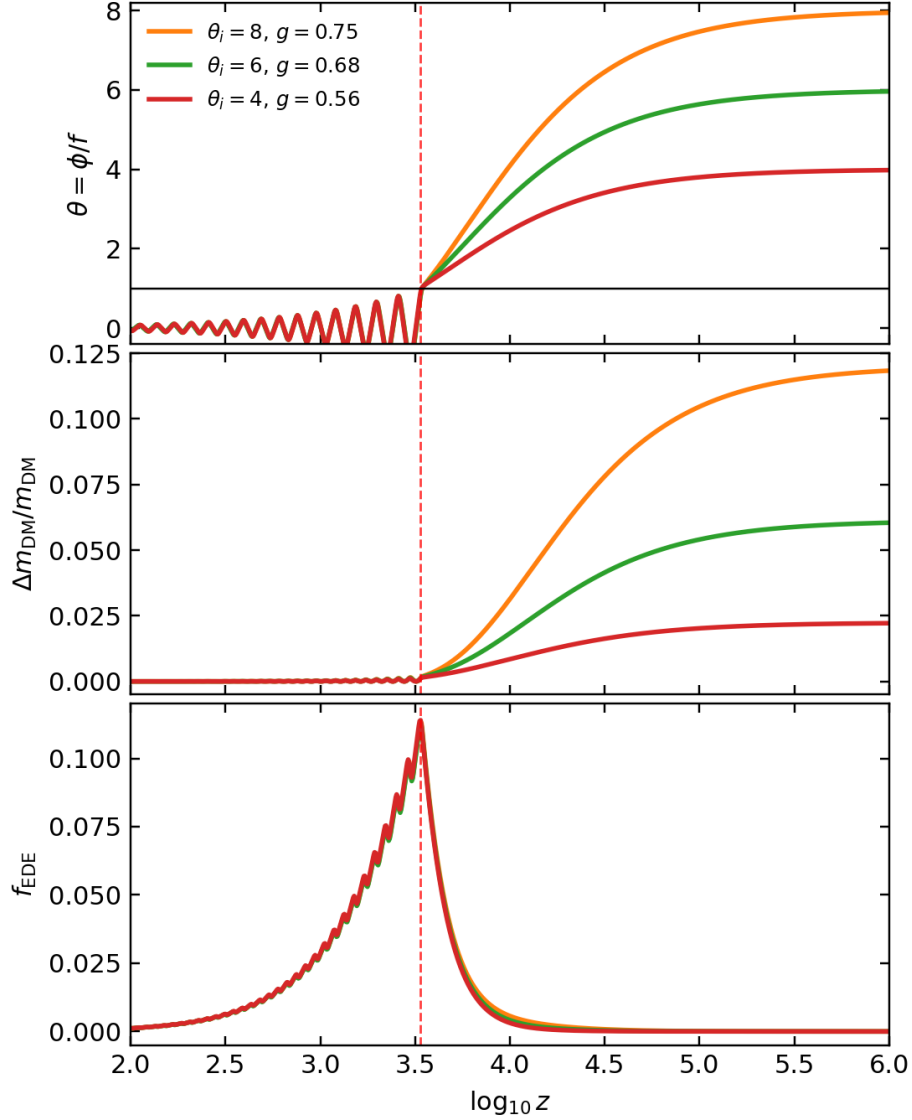


FIG. 1. Reproduction of Fig. 2 of Ref. [4] in AxiCLASS. Three initial conditions $\theta_i \in \{4, 6, 8\}$ on the tEDS plateau ($p = 8$, $f_{\text{tEDS}} = 0.05 M_{\text{Pl}}$) with coupling $g \in \{0.56, 0.68, 0.75\}$ and $\omega_{\text{IDM},0} = 0.1280 h^2$ enforced by CLASS internal shooting. Top: $\theta(z) = \phi/f_{\text{EDE}}$. Middle: DM fractional mass shift $g\phi^2$. Bottom: $f_{\text{EDE}}(z)$. The dashed red line marks the common peak redshift $z_{\text{peak}} \simeq 3380$ ($\log_{10} z = 3.53$), in good agreement with the paper's $z_c \approx 3162$ ($\log_{10} \approx 3.50$). All three cases release at the same redshift, confirming coupling-dominated timing.

The large axion decay constant $f = 1 M_{\text{Pl}}$ keeps the coupling active throughout the release. If one used a smaller f (e.g., $f = 0.05 M_{\text{Pl}}$) with the $(1 - \cos)^3$ potential, the coupling would similarly collapse at the peak.

This also reveals that the coupling fraction at the peak is not a meaningful distinction between the *trigger mechanisms*: in both cases the coupling initiates the field motion ($f_{\text{coup}}(z_{\text{trig}}) = 1$). The difference is purely geometric: whether the field trajectory passes through regions of small or large ϕ after release.

Appendix C: IDM-EDE normalization and the CLASS internal shooting fix

1. The problem with fixed Ω_Λ

The AxiCLASS input parameter `omega_ini_idm_edu` sets the IDM-EDE density at the initial integration redshift, corrected for the field-dependent mass at that epoch:

$$\omega_{\text{ini}} = \omega_{\text{IDM,target}} (1 + g\phi_i^2). \quad (\text{C1})$$

Under pure number conservation ($\phi \rightarrow 0$, no energy exchange), this would guarantee $\omega_{\text{IDM},0} = \omega_{\text{IDM,target}}$ today. However, the type-1 coupling transfers energy from the IDM sector to the scalar field as the mass decreases during the roll, reducing $\omega_{\text{IDM},0}$ below the target. For our benchmark parameters (tEDS, $\theta_i = 6$, $g = 0.68$) we find $\omega_{\text{IDM},0} \approx 0.94\omega_{\text{ini}}$, so the target is undershot by $\sim 6\%$.¹

A second problem appeared when $\Omega_\Lambda = 0.685$ was fixed as input: this over-constrains the flat-universe budget and forces $\Omega_{\text{IDM},0}$ to whatever residual is left after baryons, radiation, and Λ , regardless of `omega_ini_idm_edu`. In our early runs this gave a spurious $\Omega_{\text{cdm}} \approx 0.21$ (CLASS filling the dark-matter gap automatically) and $\Omega_{\text{IDM},0} = 0.194$, inconsistent with the paper’s setup.

2. Diagnosis of the original CLASS shooting

The AxiCLASS code originally listed `omega_ini_idm_edu` in the shooting target array, with `omega_idm_edu` as the unknown. Inspection of `source/input.c` revealed that the *target/unknown roles were swapped*: the code was configured to vary `omega_idm_edu` (the today value) while `omega_ini_idm_edu` was treated as a fixed hint for the initial guess. This is inverted relative to what the user wants. The shooting converged to a fixed point where `omega_idm_edu` = $\rho_{\text{IDM},0}/H_0^2$, but this fixed point is not controlled by the user’s desired target; it depends on the physics of the energy transfer and does not guarantee $\omega_{\text{IDM},0} = 0.1280$.

3. Fix: new shooting target `omega_idm_edu`

We corrected the shooting by adding `omega_idm_edu` (today value) as a proper shooting *target* with `omega_ini_idm_edu` as the corresponding *unknown*, at indices 14–15 of the `target_namestrings/unknown_namestrings` arrays in `source/input.c` (and two new enum values `Omega_idm_edu_today/omega_idm_edu_today` in `include/input.h`). The shooting residual is

$$f(y) = \Omega_{\text{IDM},0}^{\text{target}} - \frac{\rho_{\text{IDM}}(z=0)}{H_0^2}, \quad (\text{C2})$$

where $\Omega_{\text{IDM},0}^{\text{target}} = \omega_{\text{IDM,target}}/h^2$ is fixed by the user’s `omega_idm_edu` input throughout all iterations, and $\rho_{\text{IDM}}(z=0)$ is the ODE-integrated IDM density that depends on the trial value of $y = \omega_{\text{ini_idm_edu}}$. The initial guess is

$$y_0 = \omega_{\text{IDM,target}} \frac{1 + g\phi_i^2}{0.93}, \quad (\text{C3})$$

which accounts for the initial mass enhancement and the approximate 6% energy loss. The bracket-search Jacobian is $dy/df \approx -0.5$ (negative, reflecting that more initial IDM gives a larger today density and thus a more negative residual).

Table I shows the shooting convergence for the tEDS benchmark. The CLASS internal Newton solver finds $\omega_{\text{ini_idm_edu}} = 0.13583$ in five function evaluations, giving $\omega_{\text{IDM},0} = 0.1280 h^2$ to machine precision. All analysis scripts now use `omega_idm_edu` = 0.1280 with `do_shooting` = yes.

The corrected $z_{\text{peak}} \approx 4400$ and $\Omega_\Lambda \approx 0.726$ differ from Ref. [4] Fig. 2 (which shows $z_c \approx 3162$ at the same g). Part of this residual discrepancy likely stems from our choice of $H_0 = 72.81$ km/s/Mpc (vs. the paper’s best-fit $H_0 \approx 70.7$): a lower H_0 gives a higher Ω_Λ and shifts z_{peak} downward. The dominant remaining uncertainty is therefore the H_0 prior, not the IDM normalization.

¹ The fraction depends on the coupling parameters and potential shape; early runs showed larger discrepancies of $\sim 19\%$ before the IDM continuity equation was corrected.

Evaluation	ω_{ini}	$\omega_{\text{IDM},0}$
0 (initial guess)	0.14606	0.13705
1	0.11662	0.10942
2	0.13240	0.12417
3	0.13583	0.12800

TABLE I. CLASS internal shooting convergence for tEDS ($\theta_i = 0.3 \text{ Mpl}$, $g = 0.68$, V_0 as in text). The target is $\omega_{\text{IDM},0} = 0.1280 h^2$. Convergence is achieved in 4–5 evaluations using the Ridders bracket method.

Appendix D: Comparison of f_{EDE} and $\Omega_{\text{IDM-EDE}}$ between the two models

Figure 3 directly overlays the IDM-EDE fractional density $\Omega_{\text{IDM-EDE}}/\Omega_{\text{tot}}$ and f_{EDE} for the tEDS benchmark ($\theta_i = 6$, $g = 0.68$) and two $(1 - \cos)^3$ cases ($m/H_0 = 30$, $f = 1$, $g = 1$ and $g = 0.3$). The IDM density tracks CDM at high z and then departs after the mass shift. The EDE bumps peak at comparable redshifts, though the $(1 - \cos)^3$ bump is broader and occurs slightly later.

Figure 4 shows the natural-parameter $(1 - \cos)^3$ scan with $f = 1$, $g = 1$, and $m/H_0 \in \{1, 3, 10, 30, 100\}$. The near-degeneracy in peak timing across all m/H_0 values is a signature of coupling domination: when the coupling sets the trigger, the timing depends on g and ρ_{IDM} rather than on m .

Appendix E: Debugging history and code changes

The results in this paper required several corrections to the AxiCLASS type-1 coupling implementation. We document them here for reproducibility.

1. Corrected type-1 IDM continuity equation

The original implementation did not include the energy-exchange term due to the time-varying IDM mass. The correct equation is

$$\frac{d\rho_{\text{IDM}}}{d \ln a} = -3\rho_{\text{IDM}} + \frac{\phi'}{aH} \partial_\phi \ln m_{\text{IDM}} \rho_{\text{IDM}}, \quad (\text{E1})$$

implemented at `source/background.c` lines 3386–3388. Without the second term, energy conservation between the scalar field and IDM sectors is violated.

2. Corrected type-1 KG force

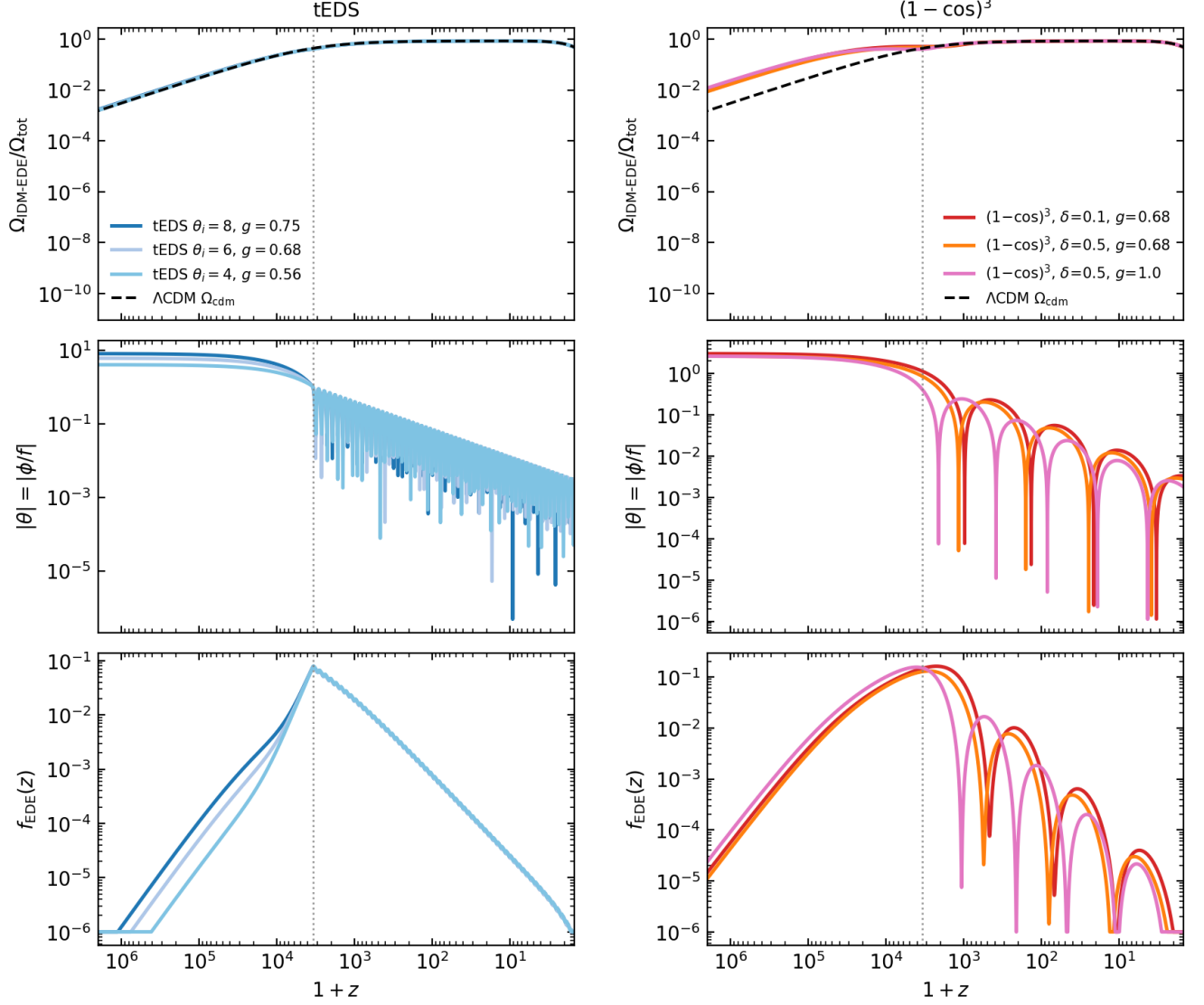
The coupled Klein-Gordon equation must include the full effective slope $V_{,\phi} + 3\rho_{\text{IDM}} \partial_\phi \ln m_{\text{IDM}}$. An earlier version used only $V_{,\phi}$ in part of the integration, leading to an inconsistency with the IDM continuity equation. The fix is at `source/background.c` line 3417.

3. Corrected polynomial second derivative

The perturbation equations require $\partial_\phi^2 \ln m$. The correct expression for $m(\phi) = m_0(1 + g\phi^2)$ is

$$\partial_\phi^2 \ln m = \frac{2g(1 - g\phi^2)}{(1 + g\phi^2)^2}, \quad (\text{E2})$$

which changes sign at $\phi = 1/\sqrt{g}$. An earlier version used a simplified expression that was only valid for $g\phi^2 \ll 1$. Fix at `source/background.c` line 4186.



Note: in both our setup and Lin et al. \(\Omega_{\text{IDM-EDE}} = \Omega_{\text{DM}}\) (\(\omega_{\text{cdm}} = 0, \omega_{\text{IDM}} = 0.1280 h^2\)).

FIG. 2. $\Omega_{\text{IDM-EDE}}/\Omega_{\text{tot}}$ (top), $|\theta|(z)$ (middle), and $f_{\text{EDE}}(z)$ (bottom) for tEDS (left) and $(1 - \cos)^3$ (right), illustrating the normalization bug. The dashed black curve is the $\Lambda\text{CDM } \Omega_{\text{cdm}}/\Omega_{\text{tot}}$ reference. Fixing $\Omega_{\Lambda} = 0.685$ produces a spurious CDM component ($\Omega_{\text{cdm}} \approx 0.21$); after the fix (`omega_cdm=0`, CLASS shooting with `omega_idm_eде`), the CDM component vanishes and $\omega_{\text{IDM},0}$ matches the target $0.1280 h^2$.

4. Corrected perturbation-level coupling source

The perturbation equations were sourced with hard-wired values for $\partial_{\phi} \ln m_{\text{IDM}}$ and $\partial_{\phi}^2 \ln m$ rather than the actual background field values, causing an inconsistency between the background and perturbation sectors for large-field runs. Fix at `source/perturbations.c` lines 10383–10388 and line 10669.

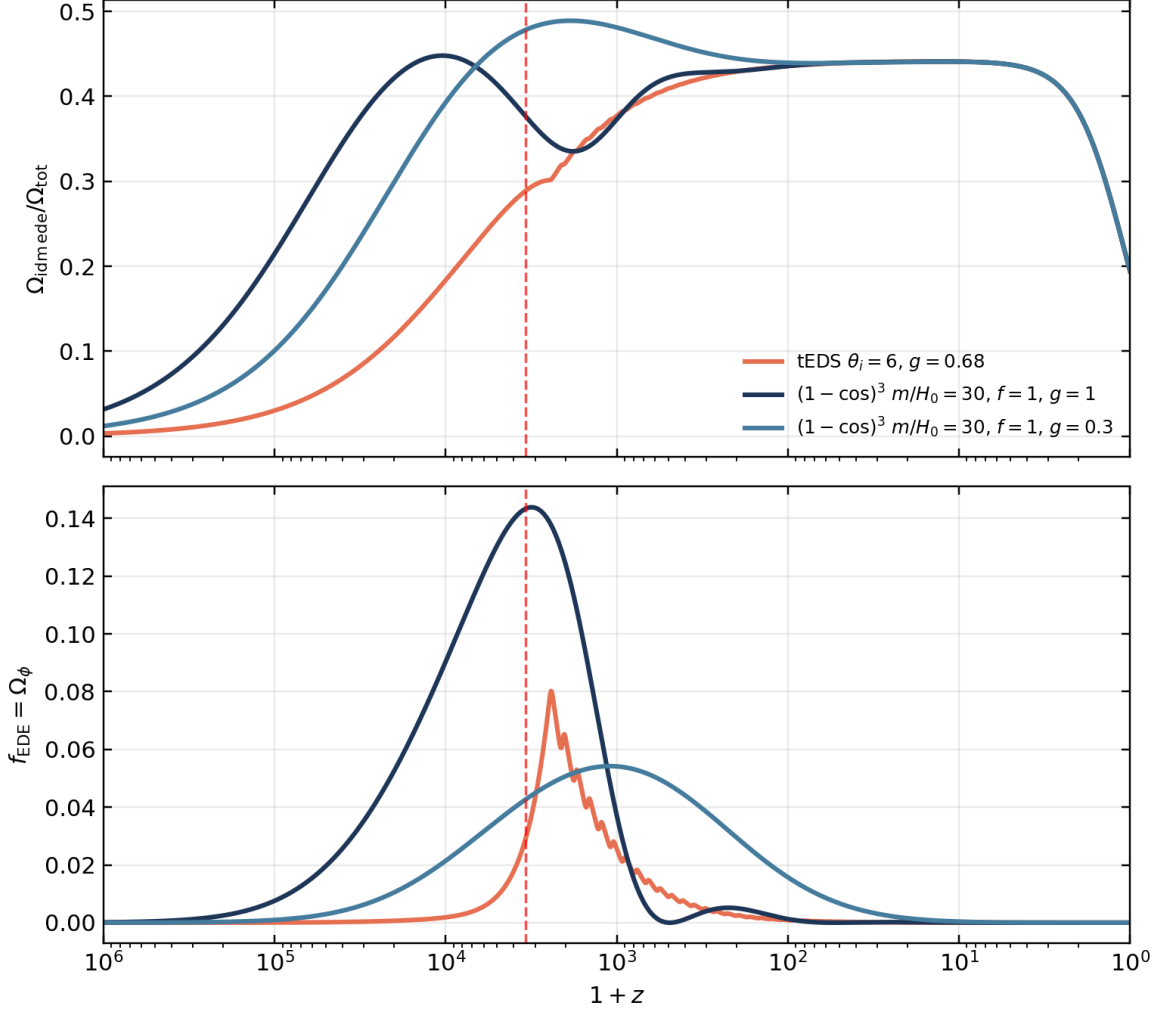


FIG. 3. Direct comparison of $\Omega_{\text{IDM-EDE}}/\Omega_{\text{tot}}$ (top) and f_{EDE} (bottom) for the tEDS benchmark ($\theta_i = 6$, $g = 0.68$) and two $(1 - \cos)^3$ benchmarks ($m/H_0 = 30$, $f = 1$, $g \in \{0.3, 1\}$). The IDM fraction tracks CDM until the coupling-driven mass shift, while the EDE bumps are qualitatively similar in timing.

5. Exponential coupling option

An exponential coupling $m(\phi) = m_0 e^{c\phi}$ (constant $\partial_\phi \ln m_{\text{IDM}} = c$) was added as `idm_ede_mass_form = exp` for comparison with the polynomial case. Polynomial coupling saturates at large ϕ (since $\partial_\phi \ln m_{\text{IDM}} \rightarrow 0$ for $g\phi^2 \gg 1$), whereas exponential coupling provides a constant force per unit IDM density. Implementation: `source/input.c` lines 3831–3846 and `source/background.c` lines 4142–4164.

6. Coupled slow-roll initial condition

Starting from $\dot{\phi}_{\text{ini}} = 0$ places the system off the overdamped branch Eq. (5), generating an unphysical transient in ρ_ϕ before the solution relaxes. Figure 5 illustrates this: the old initialization produces a visible spike in the early-time EDE density that disappears when the field relaxes to the terminal branch.

We initialize instead directly on the coupled terminal branch,

$$\phi'_{\text{ini}} = -\frac{a}{2H} (V_{,\phi} + 3\rho_{\text{IDM}} \partial_\phi \ln m_{\text{IDM}}) \Big|_{\text{ini}}. \quad (\text{E3})$$

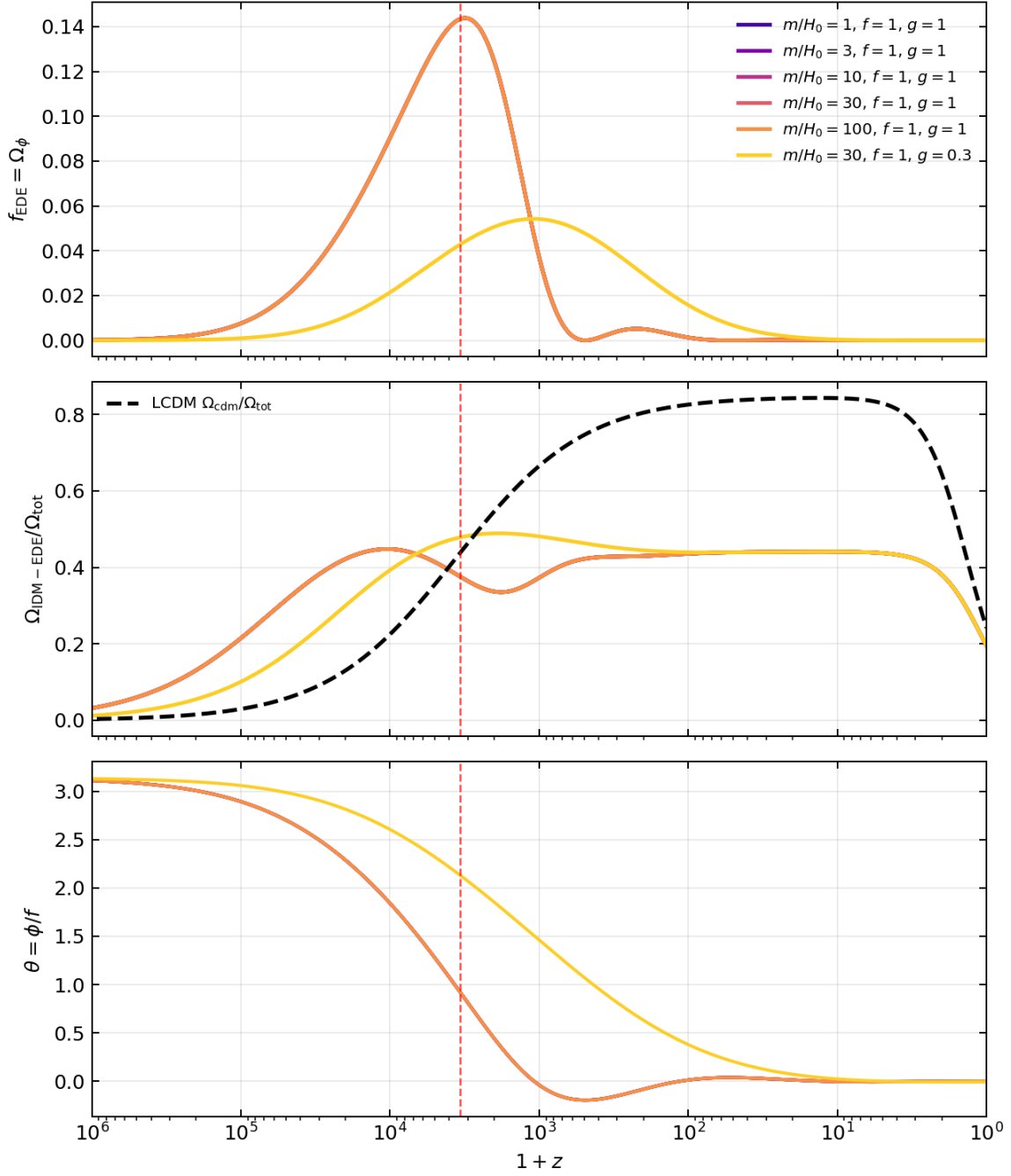


FIG. 4. $(1 - \cos)^3$ background histories for $f = 1$, $g = 1$, and $m/H_0 \in \{1, 3, 10, 30, 100\}$. The near-degeneracy in f_{EDE} peak timing across two decades in m/H_0 demonstrates coupling-dominated release: the trigger is set by the IDM dilution rate, not by m alone.

The factor of 2 (vs. the $3H$ in the cosmic-time slow-roll formula) comes from the $-2\phi'$ damping term in the $\ln a$ -form KG equation (see Appendix E 7); setting $d\phi'/d\ln a = 0$ gives exactly Eq. (E3). The late-time evolution is unchanged. Implementation: flag `coupled_scf_slowroll_ic`, enabled by default for type-1 runs; `source/background.c` lines 2966–2979 and `include/background.h` line 220.

7. Background equations

The implemented $\ln a$ -form equations are:

$$\frac{d\phi}{d\ln a} = \frac{\phi'}{aH}, \quad (\text{E4})$$

$$\frac{d\phi'}{d\ln a} = -2\phi' - \frac{a}{H}(V_{,\phi} + 3\rho_{\text{IDM}} \partial_\phi \ln m_{\text{IDM}}), \quad (\text{E5})$$

$$\frac{d\rho_{\text{IDM}}}{d\ln a} = -3\rho_{\text{IDM}} + \frac{\phi'}{aH} \partial_\phi \ln m_{\text{IDM}} \rho_{\text{IDM}}. \quad (\text{E6})$$

8. Polynomial coupling derivatives

For $m_{\text{IDM}}(\phi) = m_0(1 + g\phi^2)$:

$$\partial_\phi \ln m_{\text{IDM}} = \frac{2g\phi}{1 + g\phi^2}, \quad \partial_\phi^2 \ln m = \frac{2g(1 - g\phi^2)}{(1 + g\phi^2)^2}. \quad (\text{E7})$$

Both expressions are evaluated at the current background field value at each time step, in both the background (`source/background.c` line 4186) and perturbation (`source/perturbations.c` lines 10383–10388) equations.

9. Corrected V_0 unit conversion

The tEDS potential normalization V_0 is specified in CLASS units of $M_{\text{Pl}}^2/\text{Mpc}^2$. The conversion from physical units (eV^4) is

$$V_0^{\text{CLASS}} = V_0^{\text{eV}^4} \times \left(\frac{1 \text{ Mpc}^{-1}}{m_{\text{Pl}}} \right)^{-2} = V_0^{\text{eV}^4} \times \left(\frac{1.56 \times 10^{29} \text{ eV}}{2.435 \times 10^{27} \text{ eV}} \right)^2 \approx V_0^{\text{eV}^4} \times 4104. \quad (\text{E8})$$

An earlier version of the analysis scripts used an extra factor of $1/3$ in this conversion, motivated by the fact that `background.c` line 514 computes $\tilde{\rho}_{\text{scf}} = (K + V)/3$. This was incorrect: the factor of $1/3$ is already internal to CLASS and should not appear in the conversion of V_0 . With the spurious factor the stored V_0 was three times too small, giving $f_{\text{EDE}} \approx 4\%$ instead of the correct $\approx 8\%$ for $V_0 = 0.08 \text{ eV}^4$. The fix: remove the $/3$ from `EV4_TO_CLASS` and use $V_0^{\text{eV}^4} = 0.10$ (matching Ref. [4] Fig. 2 peak $f_{\text{EDE}} \approx 10\%$).

10. Fixed CLASS internal shooting for `omega_idm_ede`

The original `target_namestrings/unknown_namestrings` arrays in `source/input.c` had the IDM-EDE target and unknown roles reversed: `omega_ini_idm_ede` was listed as the target and `omega_idm_ede` as the unknown, causing the shooter to solve a fixed-point problem that ignored the user's desired today density.

The fix adds two new enum values `Omega_idm_ede_today/omega_idm_ede_today` (indices 14–15) to `include/input.h` (updating `_NUM_TARGETS_` from 14 to 16) and the corresponding entries in the three arrays (`target_namestrings`, `unknown_namestrings`, `target_cs`) in `source/input.c`. The initial guess and bracket-search Jacobian are set in `input.get_guess` (new case `Omega_idm_ede_today`). The shooting output function already computes the correct residual $\Omega_{\text{IDM},0}^{\text{target}} - \rho_{\text{IDM},0}/H_0^2$, which is shared with the output cases for the existing (swapped) targets. The bracket-search Jacobian sign is -0.5 (negative): increasing ω_{ini} gives a larger today density, making the residual more negative.

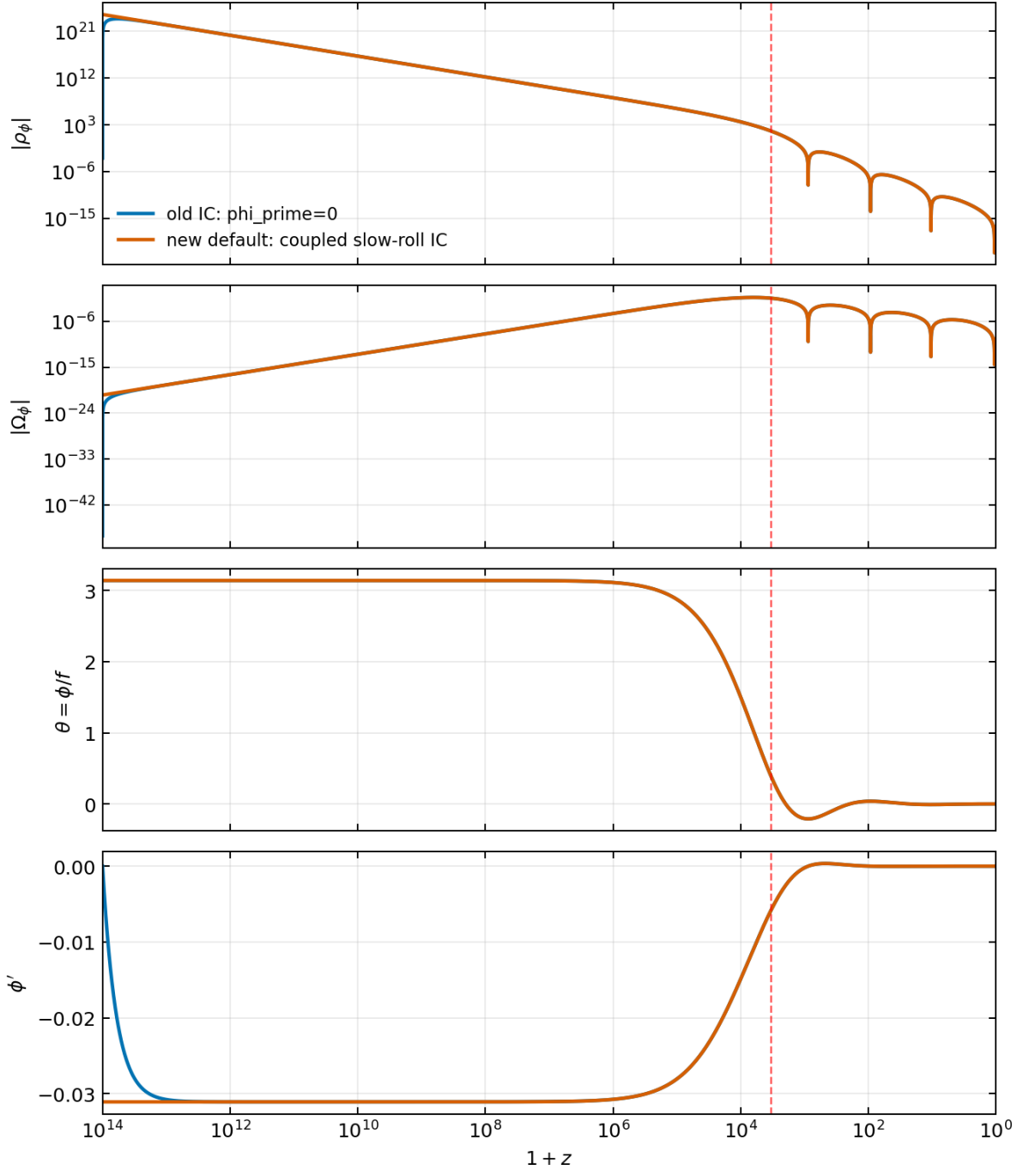
Coupled IC comparison: $f = 0.5$, $m/H_0 = 100$, $g = 1$ 

FIG. 5. Effect of the initial condition for $(f, m/H_0, g) = (0.5, 100, 1)$. The old initialization with $\phi'_{\text{ini}} = 0$ produces a transient spike in the EDE density at early times (dashed). The corrected coupled slow-roll IC starts on the overdamped branch, removing the artifact without changing the late-time history (solid).

11. Input parser: mass-form switch

The parser accepts `idm_ede_mass_form = poly` (polynomial), `exp` (exponential), `phantom/andriot/am` (phantom-sector forms). The type-1 coupling path parses the mass form and defaults at `source/input.c` lines 3821–3846 and 7707.

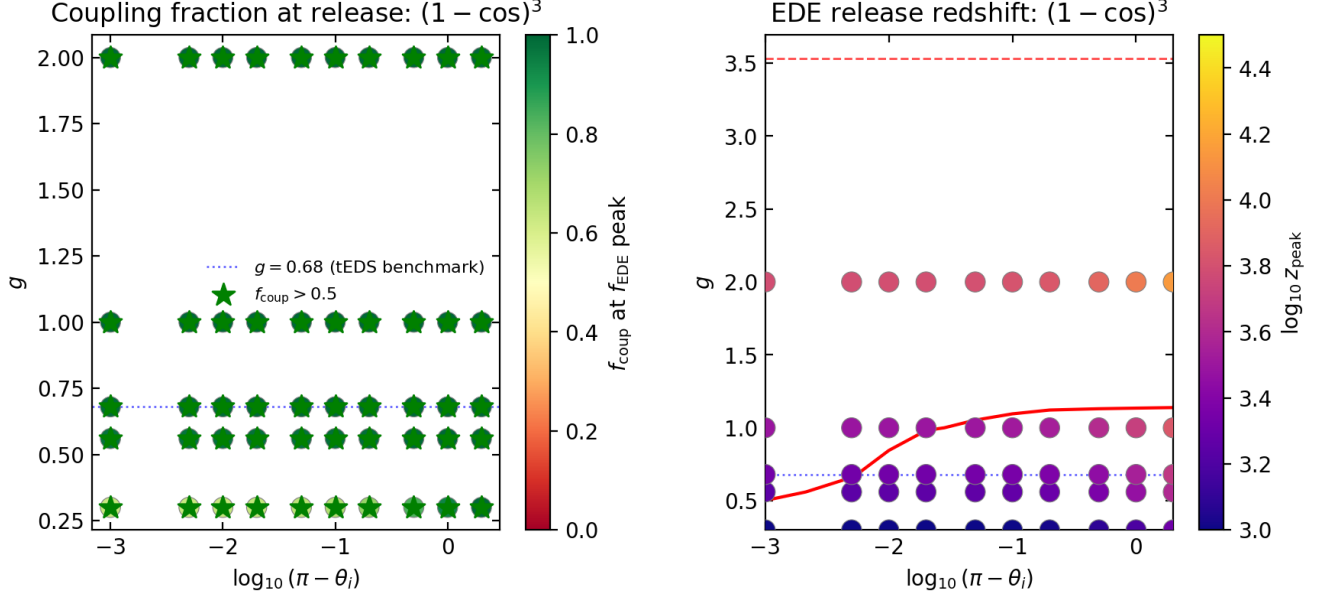


FIG. 6. Left: coupling fraction f_{coup} at the f_{EDE} peak for $(1 - \cos)^3$ in the $(\log_{10} \delta, g)$ plane ($\delta \equiv \pi - \theta_i$). Green stars mark coupling-dominated cases ($f_{\text{coup}} > 0.5$). The dashed blue line is the tEDS benchmark $g = 0.68$. Right: release redshift $\log_{10} z_{\text{peak}}$ in the same plane; the red dashed line marks z_{eq} , and the red contour traces $z_{\text{peak}} = z_{\text{eq}}$. Nearly all tested cases are coupling-dominated. The release redshift increases with g (stronger coupling earlier trigger) and is nearly independent of δ for $g \gtrsim 0.56$.

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- [1] V. Poulin, T. L. Smith, T. Karwal, and M. Kamionkowski, “Phasing in Planck Data Post Recombination and the Hubble Tension,” [arXiv:1811.04083](#).
 - [2] T. L. Smith, V. Poulin, and M. A. Amin, “Oscillating scalar fields and the Hubble tension: a resolution with novel signatures,” [arXiv:1908.06995](#).
 - [3] E. McDonough, M.-X. Lin, J. C. Hill, W. Hu, and S. Zhou, “The Early Dark Sector, the Hubble Tension, and the Swampland,” [arXiv:2112.09128](#).
 - [4] M.-X. Lin, E. McDonough, J. C. Hill, and W. Hu, “A Dark Matter Trigger for Early Dark Energy Coincidence,” [arXiv:2212.08098](#).

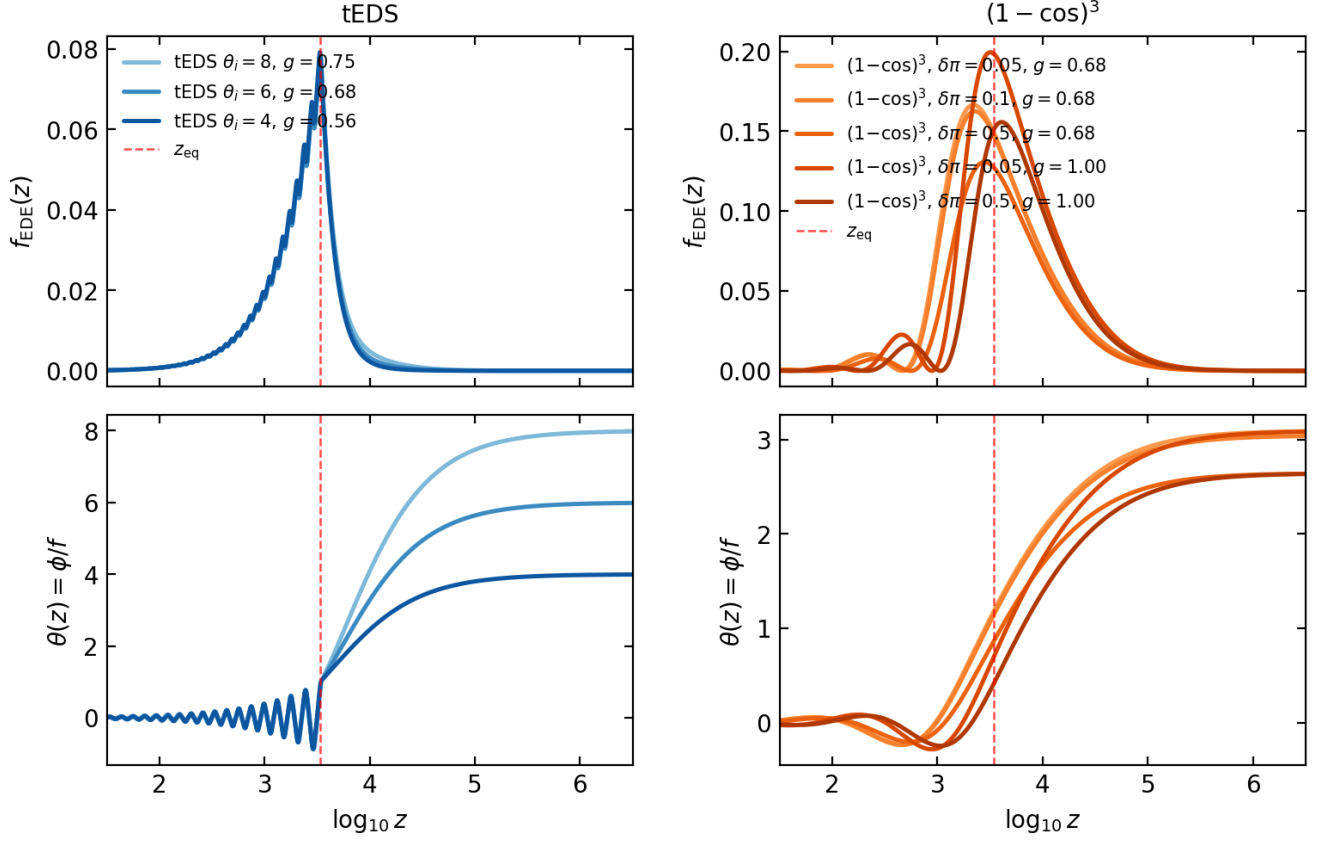


FIG. 7. EDE histories for tEDS (left column) and $(1 - \cos)^3$ (right column). Top: $f_{\text{EDE}}(z)$. Bottom: normalized field $\theta(z) = \phi/f$. The dashed vertical line marks z_{eq} . In tEDS the field rolls off the plateau and rapidly decays to $\theta \rightarrow 0$, producing a sharp f_{EDE} bump. In $(1 - \cos)^3$ the field drifts slowly away from the hilltop, producing a broader bump.

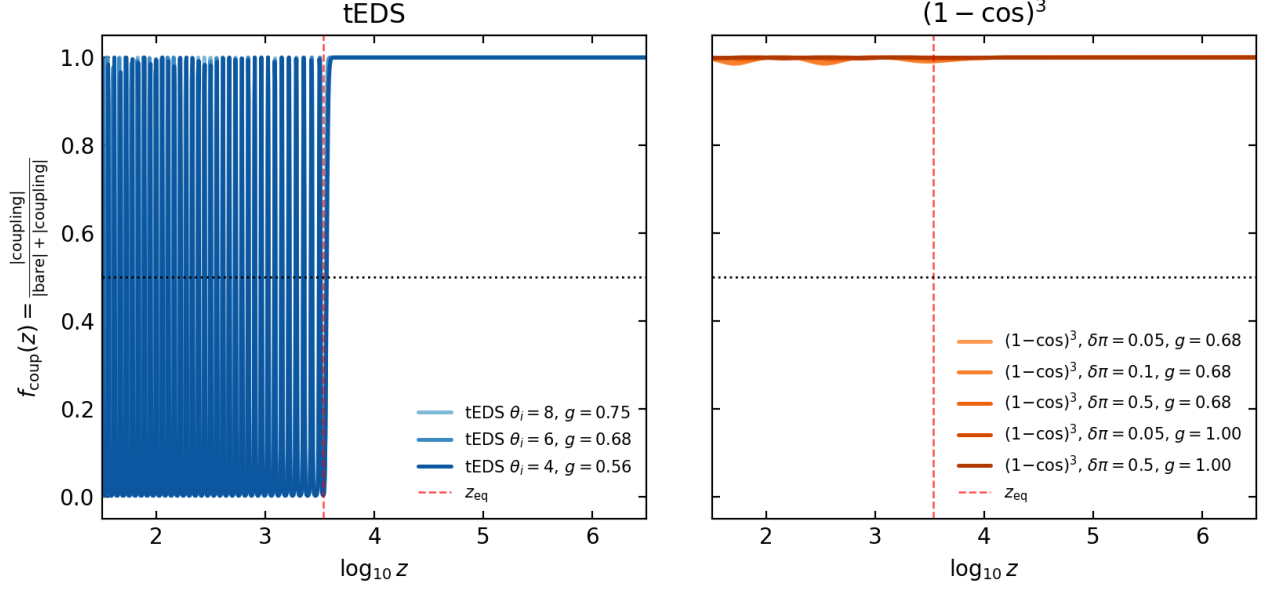


FIG. 8. Coupling fraction $f_{\text{coup}}(z)$ as a function of redshift. Left: tEDS benchmarks. Right: $(1 - \cos)^3$ with $g = 0.68$ and $\delta \in \{0.05, 0.1, 0.5\}$ and $g \in \{0.68, 1\}$. In both potentials $f_{\text{coup}} \approx 1$ at $z > z_{\text{eq}}$, confirming coupling-dominated release. For tEDS, after rolling off the plateau the field oscillates rapidly around $\phi = 0$, causing fast oscillations in f_{coup} (the ratio $|\text{coupling}|/|V'|$ alternates sign as the field passes through zero). In $(1 - \cos)^3$ the field drifts slowly away from the hilltop and f_{coup} remains ≈ 1 throughout since the bare slope is always negligible near $\theta = \pi$.

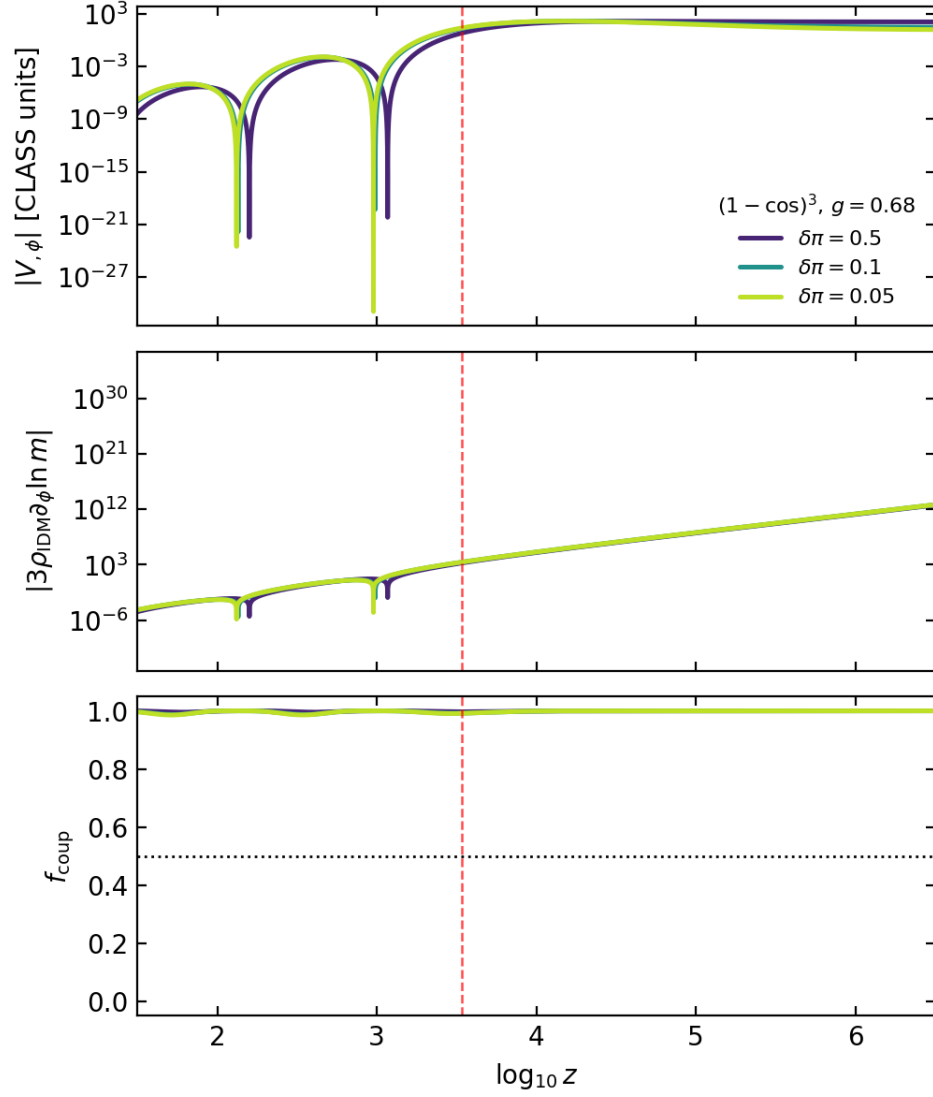


FIG. 9. Force budget for $(1 - \cos)^3$ at $g = 0.68$ and $\delta \in \{2, 0.5, 0.1, 0.01\}$. Top: bare slope $|V_\phi|$. Middle: coupling term $|3\rho_{\text{IDM}}\partial_\phi \ln m|$. Bottom: coupling fraction $f_{\text{coup}}(z)$. The coupling dominates ($f_{\text{coup}} \approx 1$) from high redshift until well past the EDE peak for all δ values. The bare slope is orders of magnitude smaller than the coupling term at $z \gtrsim 10^4$ regardless of δ .